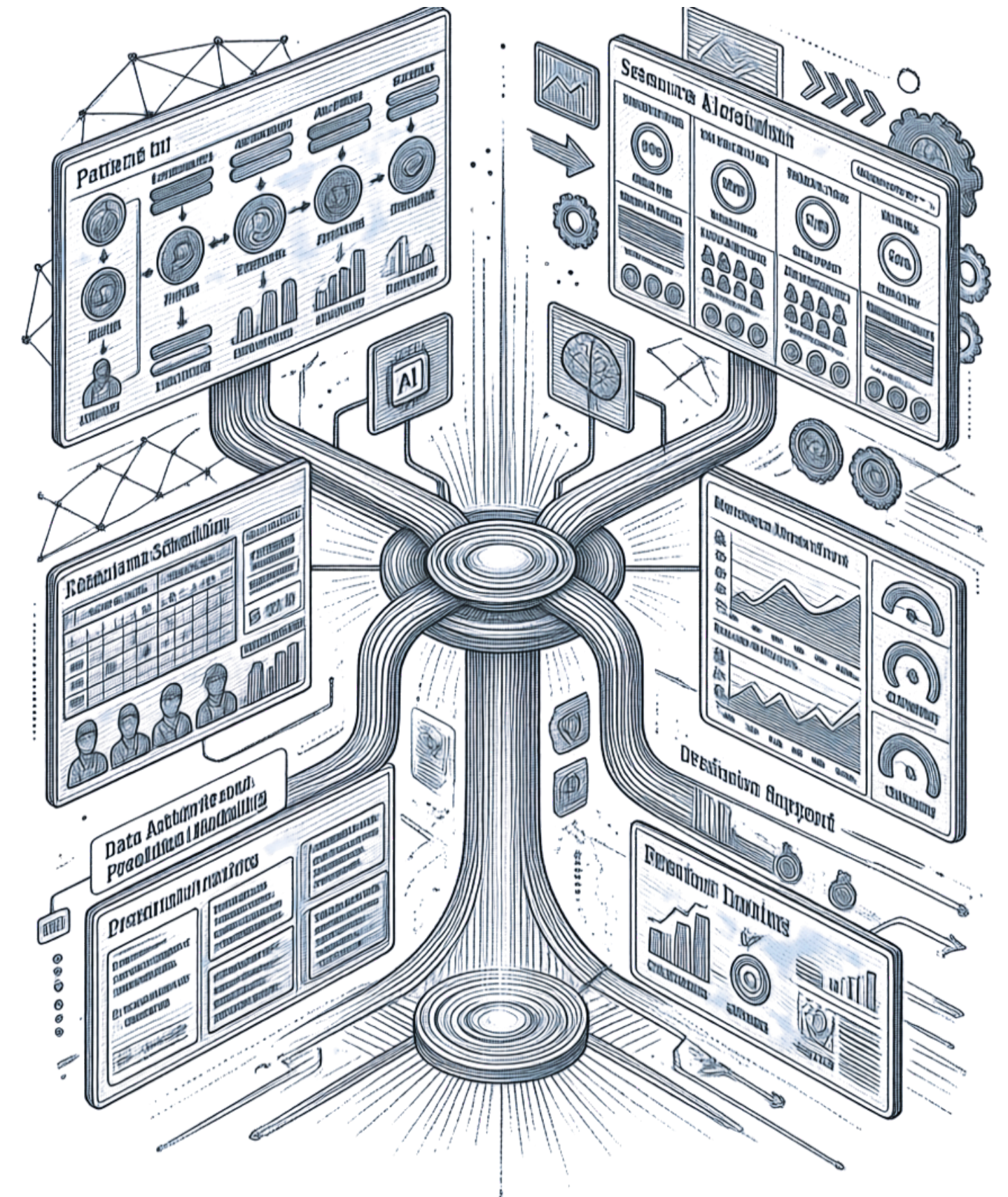


DEVELOPMENT OF A Healthcare Operations Intelligence Dashboard with Decision Analytics

Project Focus

Healthcare data management • Operational monitoring • Interactive analytics
Decision support • Alerts

GROUP 01 | Mentor: Ms. Mousami Shrivastava



02 THE HEALTHCARE DATA CHALLENGE

Why hospital data needs to become actionable intelligence

PROBLEM STATEMENT

Fragmented Data → Delayed Decisions

- Hospital information is scattered across systems and files.
- No unified view of patient flow, capacity, emergencies & resources.
- Manual analysis slows down operational decision-making.
- Bottlenecks may remain hidden until they impact service quality.

OUR SOLUTION

One Dashboard → One Operational View

- Integrate hospital data into a centralized intelligence dashboard.
- Convert raw data into reliable KPIs, trends & operational insights.
- Monitor critical hospital operations from a single interface.
- Add decision analytics, alerts & forecasting to support proactive action.

KEY ADVANTAGES

Why Our Approach Stands Out

- Single Source of Truth — One reliable operational view.
- Proactive Alerts — Flags critical conditions using thresholds.
- Forecasting-Ready — Supports future demand planning.
- Faster Decisions — Moves insight closer to real-time action.
- Reliable Architecture — Modular code with testing.

Fragmented Data → Intelligence Dashboard → Actionable Decisions

Most dashboards show what happened. Ours helps identify what requires attention.

DASHBOARD CAPABILITIES & USER VALUE

Turning raw hospital data into an understandable operational picture

KPI Monitoring

Appointments, emergency cases, patient operations, laboratory activity and operational KPIs.

Interactive Analysis

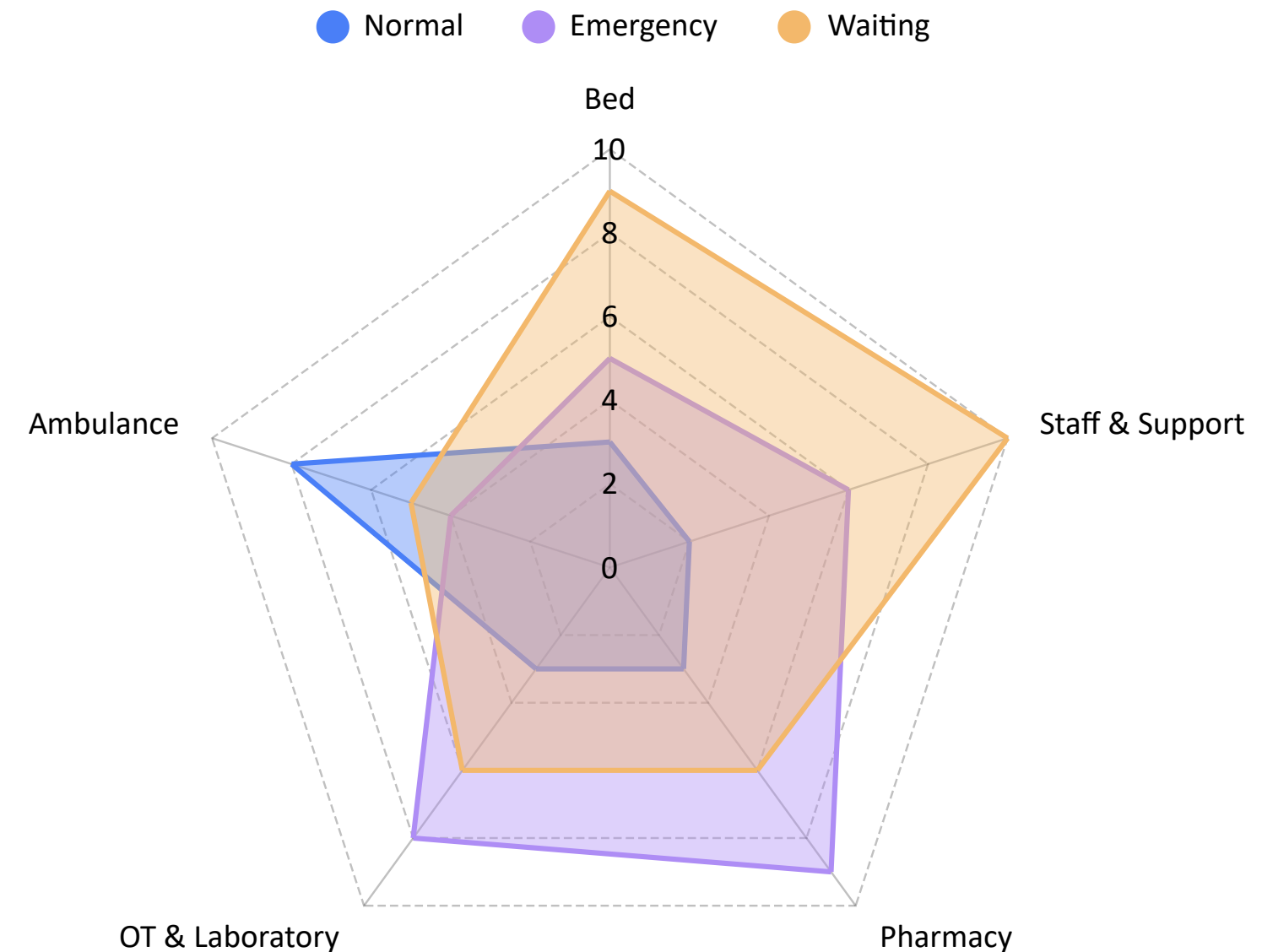
Filters, date ranges and department/category views for drill-down analysis.

Data Visualization

Bar, line and pie charts plus trends and correlations for rapid interpretation.

Decision Support

Critical-condition detection, resource-planning signals and forecasting-ready insights.



Fragmented data → Centralized dashboard → Clear insights → Proactive decisions

Development flow: data foundation → analytics engine → intelligence delivery

1 DATA FOUNDATION

Excel / Hospital datasets
OpenPyXL + Pandas
Clean • transform • validate



2 ANALYTICS ENGINE

Python + Pandas
KPI calculation • trends • correlations
threshold analysis



3 INTELLIGENCE DELIVERY

Streamlit dashboard
Decision analytics • alerts • forecasting
Management validation

COLLECT → CLEAN → TRANSFORM → ANALYZE → PREDICT → ALERT → DECIDE

Raw hospital data → reliable data layer → operational intelligence → decision support

High waiting time

Review patient flow; increase counters/staff when needed.

High bed occupancy

Plan additional capacity and accelerate discharge.

Emergency overload

Deploy additional triage/support resources.

Low ambulance response

Review routing and fleet allocation.

High OT utilization

Optimize scheduling and capacity.

KPI / Operational Signal

Compare with threshold

Condition requires attention?

Generate alert / insight

Management reviews

Take appropriate action

FUTURE SCOPE & SCALE-UP ROADMAP

Future Evolution: From Analytics Dashboard to Healthcare Intelligence Platform

01

Real-Time Data Integration

Live hospital databases & APIs

02

AI-Based Prediction

Patient, emergency, bed, staff & pharmacy demand

03

Automated Notifications

Email, SMS & mobile threshold alerts

04

Statsmodels / ML forecasting

Demand prediction, capacity planning & workload forecasting

05

Mobile & Secure Access

Role-based mobile access & stronger controls

06

Hospital System Integration

Connect with existing hospital management systems

Expected Impact • Faster visibility • Earlier intervention • Better resource utilization • Proactive planning • Scalable operations

CONCLUSION

WHY HOSPITALS SHOULD ADOPT OUR APPLICATION

- Healthcare-Focused

Built around real hospital KPIs & workflows

- Actionable Intelligence

Moves beyond charts to alerts & decisions.

- Cost-Effective

Built with widely available open-source technologies.

- Easy to Deploy

Simple Excel-based data flow for quick setup.

- Proactive Planning

Forecasting enables planning before problems occur.

- Scalable

Ready for real-time data, hospital-system integration.

HOSPITAL DATA → OPERATIONAL INTELLIGENCE → PROACTIVE DECISIONS → BETTER HEALTHCARE OPERATIONS